

Example 1. Let a and b be integers having $\gcd(a, b) = d$. Prove that $a/d \perp b/d$.

Proof. Write d as an integral combination of a and b .

$$d =$$

Next divide both sides by:

$$1 =$$

Since d is a common divisor of a and b it follows that a/d and b/d are:

Hence 1 is an integral combination of a and b . Thus, $a \perp b$.

□

Example 2. Let n be an integer. Prove that n and $n + 1$ are relatively prime.

Solution. We write 1 as an integral combination of n and $n + 1$, whence $n \perp n + 1$.

$$1 =$$

□

A. Relative Primeness and Divisibility. We establish few helpful results.

First observe that, if $a \mid c$ and $b \mid c$, then:

Let $a \perp b$ be relatively prime integers.

Let c be an integer such that $a \mid c$ and $b \mid c$. Then:

Proof. Because $a \perp b$, we may write 1 as an integral combination of a and b .

$1 =$

Next, multiply by:

Because $a \mid a$ and $b \mid c$, it follows that:

and the first summand is a multiple of ab .

Because $b \mid b$ and $a \mid c$, it follows that:

and the second summand is a multiple of ab .

Since c is an integral combo of multiples of ab , it is also a multiple of ab .

□

Euclid's Lemma. Let $a \perp b$ be relatively prime integers.

Let c be an integer such that $a \mid bc$. Then:

A lemma is a result that is primarily used to prove another result.

Proof. Because $a \perp b$, we may write 1 as an integral combination of a and b .

$1 =$

Next, multiply by:

Evidently, the first summand is a multiple of a .

Because $a \mid bc$, it follows that:

Since c is an integral combo of multiples of a , it is also a multiple of a .

□

B. **Lcms.** We list the positive common multiples of two integers, like 12 and 8.

The integers will have a least positive common multiple provided:

We call this the **least common multiple** and write: $\text{lcm}(12, 8) =$

Observe the following relationship.

$$\frac{12 \cdot 8}{\text{gcd}(12, 8)} =$$

The least common multiple is always defined to be the least **positive** common multiple.

Theorem. Let a and b be positive integers. Then:

$$\text{lcm}(a, b) =$$

Proof. For simplicity of notation, let $d = \text{gcd}(a, b)$.

First, we show that $(ab)/d$ is a positive common multiple of a and b .

It is positive because:

It is a multiple of a because we can write: $(ab)/d = (ab)/d$

where we note that b/d is an integer because:

An analogous argument shows that $(ab)/d$ is a multiple of b .

Next we show $(ab)/d$ is the **least** common multiple of a and b .

Let m be a positive common multiple of a and b .

We will show that $(ab)/d$ divides m , hence is $\leq m$.

First write the gcd d as an integral combination of a and b :

$$d =$$

Next, consider the ratio:

$$\frac{m}{(ab)/d} =$$

Because m is a common multiple of a and b , all terms are integers.

Hence, the ratio is an integer, meaning that $(ab)/d$ divides m .

□

Corollary. If m is a common multiple of a and b , then: